ثانوية المدينة التعليمية
Education City High School

1.2

Rates of Change

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Average rate of change is the slope of the secant line through two points.

2. Write one precise sentence explaining this principle: The numerator and denominator must describe changes over the same interval.

3. Write one precise sentence explaining this principle: Units are output units per input unit.

4. Write one precise sentence explaining this principle: A positive average rate means the net output change is positive; it does not guarantee the function increased everywhere.

Core

5. Find the average rate of change of $f(x) = x^2 - 3x$ on $[1, 5]$.

6. A population changes from 840 to 972 over 6 years. Interpret the average rate.

7. A graph has the same value at both interval endpoints. Must its average rate be zero? Must it be constant?

8. State what the formula $\text{AROC}_{[a,b]} = \frac{f(b)-f(a)}{b-a}$ tells you and name any required restriction.

9. State what the formula $\text{units} = \frac{\text{output units}}{\text{input units}}$ tells you and name any required restriction.

10. Create a second representation (table, graph, formula, or context) for one central relationship in Rates of Change. State its domain and justify one key feature.

Medium

11. Find the average rate of change of $f(x) = 2x^2 - 3x + 1$ on $[1, 5]$.

12. Find the average rate of change of $f(x) = x^2 + 4x - 2$ on $[-2, 3]$.

13. Find the average rate of change of $f(x) = 3x^2 + x$ on $[0, 4]$.

14. Find the average rate of change of $f(x) = -x^2 + 5x + 2$ on $[2, 6]$.

15. Find the average rate of change of $f(x) = x^2 + 1$ on $[1, 4]$.

Hard

16. A car travels 42 km in the first 0.6 h and reaches 108 km from its start at 1.5 h. Find and interpret its average rate over $[0.6, 1.5]$.

17. For $f(x) = x^3$, compare AROC on $[-2, -1]$ and $[1, 2]$.

18. Find k so the average rate of $f(x) = x^2 + kx$ on $[1, 4]$ is 9.

19. A continuous function has AROC 0 on $[0, 6]$. Give two conclusions that are valid and one tempting conclusion that is invalid.

Difficult

20. For $f(x) = x^2$, show that AROC on $[a, b]$ equals $a + b$.

21. Find all intervals of length 4 on which $f(x) = x^2 - 6x$ has average rate 2.

22. A function has AROC 4 on $[1, 3]$ and AROC -2 on $[3, 8]$. Find AROC on $[1, 8]$.

Challenge / HOT

23. Prove that if a function has the same average rate m on every interval $[a, b]$, then $f(x) = mx + c$.

24. Construct a nonconstant polynomial whose average rate on $[-3, 3]$ is 0 and whose average rate on $[0, 3]$ is 6.

AP-Style Multiple Choice

25. AROC of x^2 on $[2, 5]$ is

(A) 3
(B) 7
(C) 9
(D) 21

26. If output is meters and input is seconds, AROC units are

- **(A)** m
- **(B)** s
- **(C)** m/s
- **(D)** s/m

27. AROC is zero when
- **(A)** $f(a) = 0$
- **(B)** $f(b) = 0$
- **(C)** $f(a) = f(b)$
- **(D)** $a = b$

28. Positive AROC guarantees that
- **(A)** the function is increasing everywhere
- **(B)** $f(b) > f(a)$
- **(C)** the graph is concave up
- **(D)** all local rates are positive

29. For linear $f(x) = mx + b$, AROC on any nonzero interval is
- **(A)** b
- **(B)** m
- **(C)** m+b
- **(D)** interval length

30. If AROC on $[1, 4]$ is 5 and $f(1) = 2$, then $f(4) =$
- **(A)** 7
- **(B)** 12
- **(C)** 17
- **(D)** 22

AP-Style Free Response

31. A water tank volume is $V(t) = 600 - 24t + 0.4t^2$, liters, for $0 \leq t \leq 30$. (a) Find AROC on $[5, 15]$. (b) Interpret it. (c) Compare with AROC on $[15, 25]$. (d) What does the comparison reveal?

32. Let $f(x) = x^3 - 3x$. (a) Derive AROC on $[a, a+h]$. (b) Evaluate for $a = 1, h = 0.1$. (c) Explain what happens as h becomes small, without using derivative notation.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.